

STUDY OF POLYPHENOLS CONTENT AND DRY RESIDUE IN PHYTOSUBSTANCES FROM *ASTER NOVI-BELGII* L. HERB

Storozhuk O.,

*Student of the Faculty of Pharmacy, I. Horbachevsky Ternopil National Medical University,
Maidan Voli, 1, Ternopil, Ukraine, 46001*

Budniak L.

PhD, Associate Professor, Department of Pharmacy Management, Economics and Technology, I. Horbachevsky Ternopil National Medical University, Maidan Voli, 1, Ternopil, Ukraine, 46001 ORCID: 0000-0002-4869-1344

Scopus-Author ID 57211323941

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INTRODUCTION

The development and implementation of medicinal products based on medicinal plant materials are particularly relevant for effective and high-quality prevention and comprehensive treatment of diseases. Medicinal plants have long attracted attention because they contain biologically active substances (BAS) that give them therapeutic properties (Budnyak, Storozhuk, 2023).

As is known, in plants, BAS are found in optimal ratios. Medicinal products based on plant raw materials affect the body through a complex of BAS and trace elements. These substances act at the intracellular metabolism level, thus easily penetrating tissues – this is especially important in the treatment of chronic diseases. Another advantage of phytotherapy is the low number of side effects, even with long-term use. Special attention deserves the study of such a plant as *Aster novi-belgii* L.

Aster novi-belgii L. is a perennial herbaceous plant of the Aster family (Asteraceae). The plant originates from Eastern Canada and the Northeastern United States. Since the 17th century, *Aster novi-belgii* has been widespread in Central Europe, where it was cultivated as an ornamental plant (Ibrahim, 2006).

Species of the genus *Aster* have been widely used for millennia on the Qinghai-Tibet Plateau for detoxification and treatment of seasonal pandemic diseases (Li, 2022). In addition, they are used in traditional medicine to treat fever, colds, tonsillitis, and for snake and bee bites (Shao, 1995).

The therapeutic properties of *Aster novi-belgii* L. are associated with the presence of a complex of BAS in the plant.

Phytochemical analysis of *Aster* genus species indicates the presence of various classes of secondary metabolites, including phenylpropanoids (Liu, 2010), caffeoylquinic acid (Nugroho, 2009), and saponins (Corea, 2004). It is known that these compounds have a wide range of pharmacological properties, including hemolytic, anti-cholesterolemic, immunostimulatory, and anticarcinogenic effects (Sparg, 2004).

The herb of *Aster novi-belgii* L. contains a significant amount of phenolic compounds and essential oil, which includes 97 components (Marchyshyn, 2010). Additionally, the plant contains flavonoids and hydroxycinnamic acids.

The aim of the research was to determine the content of polyphenols and dry residue in phytosubstances from the *Aster novi-belgii* L. herb.

MATERIALS AND METHODS. Phytosubstances (dense extracts) obtained from dry plant raw materials – *Aster novi-belgii* L. herb harvested during

the mass flowering period of the plant in 2023 in the Ternopil region were the object of the research.

Obtaining phytosubstance. Dense extract was obtained by maceration with stirring. Ethanol (10% v/v, 20% v/v, 30% v/v, 40% v/v, 50% v/v, 60% v/v, 70% v/v, 80% v/v) was used for infusion. After infusion, the obtained extract was filtered and dried.

Determination of total polyphenols content

Initial solution. The aliquot of the obtained alcohol extract is placed in a 25 ml volumetric flask and added water to the mark, stirred and, if necessary, filtered.

Tested solution. 2.0 ml of the initial solution is placed in a 25 ml volumetric flask, 1.0 ml of phosphorus-molybdenum-tungsten reagent and 10.0 ml of water are added and added solution of 290 g/l sodium carbonate to the mark, stirred.

Standard solution. 50.0 mg of pyrogallol is placed in a 100 ml volumetric flask and added water to the mark, stirred. 5.0 ml of the obtained solution is placed in a 100 ml volumetric flask and added water to the mark, stirred.

Comparison solution. 2.0 ml of a standard solution of pyrogallol is placed in a 25 ml volumetric flask, 1.0 ml of phosphorus-molybdenum-tungsten reagent and 10.0 ml of water are added and added solution of 290 g/l sodium carbonate to the mark, mixed. 30 minutes later, the optical density of the tested solutions and the comparison solution are measured at wavelength 760 nm, using water as a compensatory solution.

The content of the sum of polyphenols was calculated in terms of pyrogallol, as a percentage.

The research was conducted by the BAS in five replicates. Statistical analysis of the results was carried out in accordance with the requirements of the State Pharmacopoeia of Ukraine 2.0 5.3.N.1 using Microsoft Excel 2010 software (SPU, 2015).

RESULTS AND DISCUSSION.

Determination of the total polyphenols content.

The results of the study on the influence of extractant concentration on the total polyphenols content in phytosubstances obtained from the aerial part of *Aster novi-belgii* L. are presented in Table.

In the phytosubstance obtained by maceration with stirring, using 10% ethanol (1st series), 14.76% of polyphenols were extracted.

The total polyphenols content determined in the obtained phytosubstance (series 2) was 1.03 times higher compared to the content in the phytosubstance (series 1) and amounted to 15.3%.

The quantitative content of total polyphenols in the phytosubstance (series 3) was 15.04% and was 1.02

The quantitative content of total polyphenols determined in the phytosubstance (series 6) obtained using 60% ethanol was slightly lower compared to the content of these compounds in the phytosubstance (series 5) and amounted to 11.68%.

The amount of hydroxycinnamic acids in the phytosubstance (series 7) obtained using 70% ethanol was 10.94% and was lower than the amount of these substances in the previous series of phytosubstances.

From series 8, using 80% ethanol as the extractant, 8.87% of polyphenols were extracted, which is 1.23 times less than in the previous series.

Table

Metrological characteristics of the quantitative determination results of the total polyphenols content in the phytosubstances (series 1-8)

the phyto-substances (series 1-8)									
m	n	X _i	X _{av}	S ²	S _{av}	P	t(P, n)	Confidence interval	ε _{_, %}
1	2	3	4	5	6	7	8	9	10
Series 1									
5	4	14,3097	14,76	0,1041	0,1443	0,95	2,78	14,76 ± 0,40	2,72
		15,0832							
		14,5631							
		15,0052							
		14,8614							
Series 2									
5	4	15,1517	15,30	0,0433	0,0931	0,95	2,78	15,30 ± 0,26	1,69
		15,463							
		15,5658							
		15,0851							
		15,2111							
Series 3									
5	4	14,9284	15,04	0,0079	0,0398	0,95	2,78	15,04 ± 0,11	0,74
		15,174							
		15,0387							
		15,0036							
		15,0304							
Series 4									
5	4	14,5632	14,25	0,0553	0,1052	0,95	2,78	14,25 ± 0,29	2,05
		14,2854							
		13,9872							
		14,0462							
		14,3564							
Series 5									
5	4	13,0152	12,79	0,0382	0,0874	0,95	2,78	12,79 ± 0,24	1,90
		12,7951							
		12,9437							
		12,5349							
		12,6746							
Series 6									
5	4	11,2491	11,68	0,0831	0,1289	0,95	2,78	11,68 ± 0,36	3,07
		12,0037							
		11,7462							
		11,5624							
		11,8341							
Series 7									
5	4	11,2106	10,94	0,0393	0,0887	0,95	2,78	10,94 ± 0,25	2,25
		10,6773							
		10,8571							
		11,0304							
		10,9412							
Series 8									
5	4	8,6743	8,87	0,0810	0,1273	0,95	2,78	8,87 ± 0,35	3,99
		9,1831							
		9,1658							
		8,5647							
		8,7863							

Determination of dry residue.

The determination of dry residue in the dense extract was carried out according to the State Pharmacopoeia of Ukraine 2.8.16.

1.0 g (exact weight) of the phyto-substance was placed in a weighed flask, approximately 3 cm in height and 5 cm in diameter. It was evaporated to dryness on a water bath and dried in a drying cabinet at a temperature of 100 °C to 105 °C for 3 hours. The flask was

cooled in an exsiccator over phosphorus (V) oxide at room temperature for 30 minutes and then weighed.

The results of the investigation of dry residue in the phytosubstances obtained from the aerial part of *Aster novi-belgii* L. are presented in Figure.

Analyzing the dry residue content in all series, it can be concluded that the highest dry residue was obtained when using 60% ethanol. The dry residue in the 6th series amounted to 86.87%.

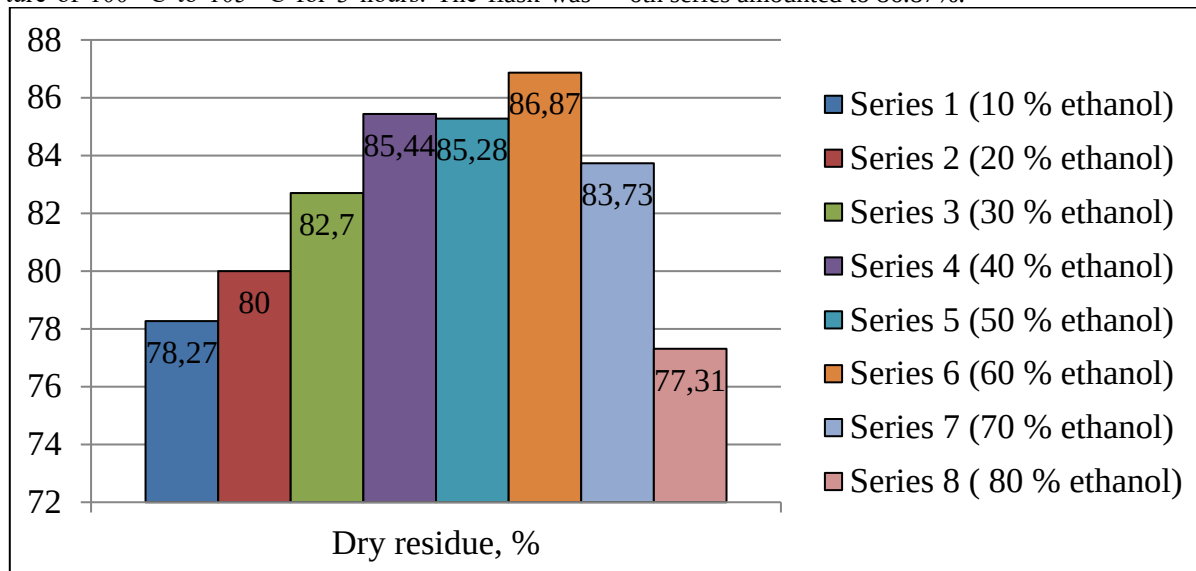


Figure. Diagram of determination of dry residue in phytosubstances from *Aster novi-belgii* L. aerial parts

Conclusions.

1. Phytosubstances from the aerial parts of *Aster novi-belgii* L. were obtained using the maceration method with stirring.

2. The quantitative content of total polyphenols was determined by spectrophotometric method based on pyrogallol. The highest amount of investigated compounds was extracted when using 20% ethanol as the extractant.

3. The gravimetric method was employed to determine the dry residue of the obtained phytosubstances. The highest dry residue, namely 86.87%, was obtained from series 6, obtained using 60% ethanol as the extractant.

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